

Convention: Throughout this discussion a feasible direction $\mathbf{d}$ at a point is by definition taken to be a nonzero vector, although there is no significant harm even if assumed otherwise. Sometimes I may have forgotten to explicitly write it.

Notation: $\nabla^2 f(\mathbf{x}) = H(\mathbf{x})$ is the Hessian matrix of f at $\mathbf{x}$.

$\nabla f(\mathbf{x})$ is the gradient **row** vector (or the vector of partial derivatives written as a row vector).

Nonlinear Programming

Let f be a real valued function defined on $\Omega \subseteq \mathbb{R}^n$.

The problem is to minimize or maximize $f(\mathbf{x})$ subject to $\mathbf{x} \in \Omega$, where f need not be a linear function.

Throughout the discussion we will assume that $\Omega \subseteq \mathbb{R}^n$, for some n .

Definition 1: A point (or an element) $\mathbf{x}^* \in \Omega$ is called a local minimum (maximum) of f if there exists an $\epsilon > 0$, such that

$\mathbf{x} \in \Omega$ and $\|\mathbf{x} - \mathbf{x}^*\| < \epsilon$ implies $f(\mathbf{x}^*) \leq f(\mathbf{x})$ ($f(\mathbf{x}^*) \geq f(\mathbf{x})$).

Definition 2: A point $\mathbf{x}^* \in \Omega$ is called a global minimum (maximum) of f if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ ($f(\mathbf{x}^*) \geq f(\mathbf{x})$) for all $\mathbf{x} \in \Omega$.

Definition 3: A vector $\mathbf{d} \in \mathbb{R}^n$, $\mathbf{d} \neq \mathbf{0}$ is said to be a feasible direction at $\mathbf{x}^* \in \Omega$, if there exists a $c > 0$ such that for all t , $0 \leq t \leq c$, $\mathbf{x}^* + t\mathbf{d} \in \Omega$.

Example 1: Let $\Omega = \{[x_1, x_2]^T : x_1 \geq 0, x_2 \geq 0\}$.

At $[0, 0]^T$ if $\mathbf{d} = [d_1, d_2]^T$ is a feasible direction then $d_1 \geq 0$ and $d_2 \geq 0$.

At $[0, \frac{1}{2}]^T$ if $\mathbf{d}$ is a feasible direction then $d_1 \geq 0$ but d_2 can be any real number.

At $[\frac{1}{2}, 0]^T$ if $\mathbf{d}$ is a feasible direction then $d_2 \geq 0$ but d_1 can be any real number.

At $[\frac{1}{2}, \frac{1}{2}]^T$ any $\mathbf{d} \in \mathbb{R}^2$ will be a feasible direction.

Remark 1: If $\mathbf{x}^*$ is an interior point of Ω then any $\mathbf{d} \in \mathbb{R}^n$, $\mathbf{d} \neq \mathbf{0}$ is a feasible direction at $\mathbf{x}^*$.

First order necessary conditions for a point to be a local minimum

The results obtained in this section is based on **first order approximation** of the function f near the local minimum point $\mathbf{x}^*$.

Throughout this discussion we will assume $\Omega \subseteq \mathbb{R}^n$ and $\mathbf{x}^*, \mathbf{d}$ are elements of $\mathbb{R}^n$ for some $n \in \mathbb{N}$.

Theorem 1: Let $f : \Omega \rightarrow \mathbb{R}$ be a **continuously differentiable function** (that is, the first order partial derivatives of f exists and are continuous as functions from Ω to $\mathbb{R}$). If $\mathbf{x}^*$ is a **local minimum point** then for any **feasible direction** $\mathbf{d}$ at $\mathbf{x}^*$,

$$\nabla f(\mathbf{x}^*)\mathbf{d} \geq 0,$$

where $\nabla f(\mathbf{x}^*)$, the gradient vector of f at $\mathbf{x}^*$ is written as a row vector (the components of $\nabla f(\mathbf{x}^*)$ are the first order partial derivatives of f at $\mathbf{x}^*$) and $\mathbf{d} \in \mathbb{R}^n$ is a column vector.

Proof: Let $\mathbf{x}^*$ be a local minimum and let $\mathbf{d}$ be a feasible direction at $\mathbf{x}^*$.

Let $g(t) = f(\mathbf{x}(t))$, where $\mathbf{x}(t) = \mathbf{x}^* + t\mathbf{d}$.

Since f is continuously differentiable throughout Ω and $\mathbf{d}$ is a **feasible direction** at $\mathbf{x}^*$

$\lim_{h \rightarrow 0} \frac{f(\mathbf{x}^* + h\mathbf{d}) - f(\mathbf{x}^*)}{h}$ exists.

Since $g(h) - g(0) = f(\mathbf{x}^* + h\mathbf{d}) - f(\mathbf{x}^*)$, $\lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h}$ also exists and

$$g'(0) = \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} = \lim_{h \rightarrow 0} \frac{f(\mathbf{x}^* + h\mathbf{d}) - f(\mathbf{x}^*)}{h} = \nabla f(\mathbf{x}^*)\mathbf{d}.$$

Since $g'(0)$ exists and we look at the first order Taylor's approximation of g around $t = 0$,

$$g(t) = g(0) + tg'(0) + o(t), \text{ where } \lim_{t \rightarrow 0} \frac{o(t)}{t} = 0. \quad (**)$$

If we take $0 < t \leq c$, then $(**)$ gives,

$$f(\mathbf{x}^* + t\mathbf{d}) = f(\mathbf{x}^*) + t\nabla f(\mathbf{x}^*)\mathbf{d} + o(t).$$

Since for t sufficiently small,

$$|t\nabla f(\mathbf{x}^*)\mathbf{d}| \geq |o(t)| \text{ (provided } \nabla f(\mathbf{x}^*)\mathbf{d} \neq 0),$$

if $\nabla f(\mathbf{x}^*)\mathbf{d} < 0$, $f(\mathbf{x}^* + t\mathbf{d}) < f(\mathbf{x}^*)$ for all $t > 0$ sufficiently small, which contradicts that $\mathbf{x}^*$ minimizes f locally.

Theorem 2 : Let $f : \Omega \rightarrow \mathbb{R}$ be a **continuously differentiable** function. Let $\mathbf{x}^*$ be an **interior point** of Ω . If $\mathbf{x}^*$ is a **local minimum point** of f then

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

Proof: Follows from **Theorem 1**, by taking $\mathbf{d} = -(\nabla f(\mathbf{x}^*))^T$. Since $\mathbf{x}^*$ is an interior point we know that every $\mathbf{d} \in \mathbb{R}^n, \mathbf{d} \neq \mathbf{0}$ is a feasible direction at $\mathbf{x}^*$.

Second order necessary conditions for a point to be a local minimum

The following conditions are obtained by considering second order approximation of the function f near the local minimum point $\mathbf{x}^*$.

Theorem 3: Let $f : \Omega \rightarrow \mathbb{R}$ be **twice continuously differentiable** (that is all the second order partial derivatives of f (given by $\frac{\partial^2 f}{\partial x_j \partial x_i}$) exists and are continuous as functions from Ω to $\mathbb{R}$).

If $\mathbf{x}^*$ is a **local minimum** of f then for any **feasible direction** $\mathbf{d}$ at $\mathbf{x}^*$

1. $\nabla f(\mathbf{x}^*)\mathbf{d} \geq 0$.
2. If $\nabla f(\mathbf{x}^*)\mathbf{d} = 0$, then $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} \geq 0$.

Note: The matrix $\nabla^2 f$ (also denoted by H) is called the **Hessian matrix** of f ,

$(\nabla^2 f(\mathbf{x}^*))_{ij} = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}^*) = \left[\frac{\partial^2 f}{\partial x_j \partial x_i} \right]_{\mathbf{x}^*}$. Since f **twice continuously differentiable** the **Hessian matrix** $\nabla^2 f$ is a symmetric matrix for all $\mathbf{x} \in \Omega$.

Proof: Let $\mathbf{d}$ be a **feasible direction** at $\mathbf{x}^*$. That $\mathbf{x}^*$ satisfies condition 1 is already shown in Theorem 1.

As before, take $g(t) = f(\mathbf{x}(t))$, where $\mathbf{x}(t) = \mathbf{x}^* + t\mathbf{d}$.

Since f is **twice continuously differentiable** $g''(0)$ exists. Also the **second order Taylor's approximation** of g around $t = 0$ gives,

$$g(t) = g(0) + tg'(0) + \frac{t^2}{2!}g''(0) + o(t^2), \quad (**)$$

Since $g'(t) = (\nabla f(\mathbf{x}^* + t\mathbf{d}))\mathbf{d} = \sum_i \left(\frac{\partial f}{\partial x_i} \right)(\mathbf{x}^* + t\mathbf{d})d_i$,

$$g'(t) = \sum_i h_i(t)d_i,$$

$$\text{where } h_i(t) = \frac{\partial f}{\partial x_i}(\mathbf{x}^* + t\mathbf{d}).$$

$$\text{Hence } g''(t) = \sum_i h'_i(t)d_i,$$

$$\text{where } h'_i(t) = \left(\left(\nabla \frac{\partial f}{\partial x_i} \right)(\mathbf{x}^* + t\mathbf{d}) \right) \mathbf{d} = \sum_j \frac{\partial}{\partial x_j} \left(\frac{\partial f}{\partial x_i} \right)(\mathbf{x}^* + t\mathbf{d})d_j = \sum_j \left(\frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}^* + t\mathbf{d}) \right) d_j.$$

$$\text{Hence } g''(t) = \sum_i h'_i(t)d_i = \sum_i \left(\sum_j \left(\frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}^* + t\mathbf{d}) \right) d_j \right) d_i,$$

$$\text{Hence } g''(0) = \sum_i \left(\sum_j \left(\frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}^*) \right) d_j \right) d_i = \mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d},$$

where $\nabla^2 f(\mathbf{x}^*) (= H(\mathbf{x}^*))$ is an $n \times n$ matrix whose (i, j) th entry is given by $\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}^*)$.

Note that because we have assumed f to be **twice continuously differentiable** the matrix $\nabla^2 f(\mathbf{x}^*)$ is a symmetric matrix.

Again from $(**)$ we get,

$$f(\mathbf{x}^* + t\mathbf{d}) = f(\mathbf{x}^*) + t\nabla f(\mathbf{x}^*)\mathbf{d} + \frac{t^2}{2}\mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} + o(t^2).$$

Since for sufficiently small t ,

$$|\frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d}| \geq |o(t^2)|, \text{ (provided } \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \neq 0)$$

hence if $\nabla f(\mathbf{x}^*) \mathbf{d} = 0$ and $\mathbf{x}^*$ is a **local minimum** then it should satisfy the condition $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0$ if $\nabla f(\mathbf{x}^*) \mathbf{d} = 0$.

Theorem 4: Let $f : \Omega \rightarrow \mathbb{R}$ be a twice continuously differentiable function and let $\mathbf{x}^*$ be an interior point of Ω . If $\mathbf{x}^*$ is a local minimum of f then

1. $\nabla f(\mathbf{x}^*) = \mathbf{0}$.
2. $\nabla^2 f(\mathbf{x}^*)$ is positive semidefinite(defined later).

Proof: Follows from the previous theorem and the fact that for an **interior point** of Ω , every nonzero vector $\mathbf{d} \in \mathbb{R}^n$ is a feasible direction.

Hence from **Theorem 3**, $\nabla f(\mathbf{x}^*)(-\nabla f(\mathbf{x}^*))^T \geq 0$, which implies $\nabla f(\mathbf{x}^*) = \mathbf{0}$.

Applying the second order conclusion of **Theorem 3**, we get $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq 0$ for all $\mathbf{d} \in \mathbb{R}^n$.

Definition: A real symmetric matrix A is said to be positive semidefinite (negative semidefinite) if $\mathbf{x}^T A \mathbf{x} \geq 0$ ($\mathbf{x}^T A \mathbf{x} \leq 0$) for all $\mathbf{x} \in \mathbb{R}^n$.

Definition: A real symmetric matrix A is said to be positive definite (negative definite) if $\mathbf{x}^T A \mathbf{x} > 0$ ($\mathbf{x}^T A \mathbf{x} < 0$) for all **nonzero** vectors $\mathbf{x} \in \mathbb{R}^n$.

Remark: Note that in general a matrix satisfying the condition $\mathbf{x}^T A \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$ need not be symmetric for example $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Theorem: If A is a symmetric, $n \times n$, real matrix then the following statements are equivalent:

1. A is positive semidefinite.
2. All eigenvalues of A are nonnegative.
3. All principal minors of A are nonnegative.

Definition: λ is called an eigenvalue of an $n \times n$ matrix A if there exists an $\mathbf{x} \in \mathbb{C}^n, \mathbf{x} \neq \mathbf{0}$ (that is atleast one component of $\mathbf{x}$ is nonzero) such that $A\mathbf{x} = \lambda\mathbf{x}$.

For example the $\mathbf{0}$ matrix has all n eigenvalues equal to 0, the identity matrix I_n has all n eigenvalues equal to 1 and for an upper triangular matrix the diagonal entries are its eigenvalues.

Definition: If A is an $n \times n$ matrix and $\alpha \subseteq \{1, \dots, n\}, \beta \subseteq \{1, \dots, n\}$ then $A[\alpha, \beta]$ is the (sub)matrix obtained from A by deleting all the rows of A which do not belong to α and by deleting all the columns of A which do not belong to β .

If $\alpha = \beta$ then $A[\alpha, \alpha]$ is called a principal submatrix of A and $\det A[\alpha, \alpha]$ is called a principal minor of A .

For example if $\alpha = \beta = \{i\}$ where $i \in \{1, \dots, n\}$ then $A[\alpha, \alpha] = [a_{ii}]$ and $\det A[\alpha, \alpha] = a_{ii}$, the i th diagonal entry.

If $\alpha = \beta = \{i, j\}$ where $i, j \in \{1, \dots, n\}$ and $i < j$ then $A[\alpha, \alpha] = \begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}$.

If $\alpha = \{1, \dots, n\}$ then $A[\alpha, \alpha] = A$ and $\det A[\alpha, \alpha] = \det(A)$.

Remark: A nonsingular (nonzero determinant) positive semidefinite matrix is positive definite.

In the following examples there is a slight abuse of notation. Instead of writing $f([x_1, x_2]^T)$, to avoid cumbersome notation, I have written it as $f(x_1, x_2)$.

Example 1 revisited: Consider the following problem:

Minimize $f(x_1, x_2) = x_1^3 - x_1^2 x_2 + 2x_2^2$

subject to $x_1 \geq 0, x_2 \geq 0$, hence $\Omega = \{[x_1, x_2]^T : x_1 \geq 0, x_2 \geq 0\}$.

To search for points which might be local minimum points of f we initially search for points in Ω which satisfy first order necessary conditions for a local minimum. The points which obviously satisfy first order necessary conditions for a local minimum are the points at which the gradient vector vanishes.

Hence we try to find $\mathbf{x}$ at which $\nabla f(\mathbf{x}) = 0$.

$\nabla f(\mathbf{x}) = (3x_1^2 - 2x_1 x_2, -x_1^2 + 4x_2) = [0, 0]$ has two solutions $x_1 = 0, x_2 = 0$ and $x_1 = 6, x_2 = 9$.

Here $[6, 9]^T$ an interior point of the feasible region $\Omega = \{[x_1, x_2]^T : x_1 \geq 0, x_2 \geq 0\}$ satisfies the first order necessary conditions for a local minimum.

But since $\nabla^2 f(\mathbf{x}) = \begin{pmatrix} 6x_1 - 2x_2 & -2x_1 \\ -2x_1 & 4 \end{pmatrix}$.

At $\mathbf{x}^* = [6, 9]^T$, $\nabla^2 f(\mathbf{x}^*) = \begin{pmatrix} 18 & -12 \\ -12 & 4 \end{pmatrix}$ is not positive semidefinite.

Hence $\mathbf{x}^* = [6, 9]^T$ is not a local minimum point of f .

Hence the first order necessary conditions are **necessary** but **not sufficient** for a point to be a local minimum.

At $\mathbf{x}^* = [0, 0]^T$ a $\mathbf{d} \neq \mathbf{0}$ is a feasible direction if and only if $d_1 \geq 0$ and $d_2 \geq 0$.

Since $\nabla f(\mathbf{x}^*) = \mathbf{0}$, $\nabla f(\mathbf{x}^*)\mathbf{d} = 0$ for all $\mathbf{d}$.

Since $\nabla^2 f(\mathbf{x}^*) = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$, $\mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} = 4d_2^2 \geq 0$ for all $\mathbf{d}$.

Since $[0, 0]^T$ satisfies both the first and second order necessary conditions, we can only say that $\mathbf{x}^* = [0, 0]^T$ can be a candidate for local minimum, but since these are only necessary conditions we cannot conclude from previous calculations that $[0, 0]^T$ is indeed a local minimum of f in Ω .

Exercise: Check that $[0, 0]^T$ is a local minimum point of f in **Example 1**.

Solution: Note that $f(x_1, x_2) = x_1^3 - x_1^2 x_2 + 2x_2^2 = x_1^2(x_1 - x_2) + 2x_2^2$ can take values < 0 only when either $x_2 = 0$ and $x_1 < 0$ or when $x_2 \neq 0$ and $x_2 > x_1$. But $[-1, 0]^T$ is not a feasible direction at $\mathbf{x}^* = [0, 0]^T$ for $\Omega = \{[x_1, x_2]^T : x_1 \geq 0, x_2 \geq 0\}$.

For $x_2 \neq 0$ and $x_2 > x_1$ sufficiently small, that is $|x_1|, |x_2| < 1$, clearly $x_1^2(x_1 - x_2) + 2x_2^2 \geq 0$. Hence $[0, 0]^T$ is a local minimum of f .

It is quite clear that for any f , of the form $f(x_1, x_2) = x_1^3 - x_1^2 x_2 + cx_2^2, c > 0$, $\mathbf{x}^* = [0, 0]^T$ will be a local minimum point of f for the domain given in **Example 1**.

However if we take $c = 0$ in the above expression, then $f(x_1, x_2) = x_1^3 - x_1^2 x_2$, then one can check that $\nabla^2 f(\mathbf{x}^*) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, is a positive semidefinite matrix, but $\mathbf{x}^* = [0, 0]^T$ is not a local optimum point.

Hence the **first order** and the **second order** necessary conditions are **necessary** but **not sufficient** for a point $\mathbf{x}^*$ to be a **local minimum**.

Exercise: Will $[0, 0]^T$ be a local minimum point of f given in **Example 1** if $\Omega = \mathbb{R}^2$?

Example 2 : Consider the following problem:

Minimize $f(x_1, x_2) = x_1^2 - x_1 + x_2 + x_1 x_2$

subject to $x_1 \geq 0, x_2 \geq 0$.

f is a twice continuously differentiable function.

At $[\frac{1}{2}, 0]^T$, $\frac{\partial f}{\partial x_1} = 2x_1 - 1 + x_2 = 0$

$$\frac{\partial f}{\partial x_2} = 1 + x_1 = \frac{3}{2}.$$

If $\mathbf{d}$ is a feasible direction at $[\frac{1}{2}, 0]^T$, then d_2 has to be nonnegative.

Hence $\nabla f(\mathbf{x})|_{[\frac{1}{2}, 0]^T} \mathbf{d} = \frac{3}{2}d_2 \geq 0$ for any feasible direction $\mathbf{d}$.

Hence the first order necessary conditions for $\mathbf{x}^* = [\frac{1}{2}, 0]^T$ to be a locally minimum point is satisfied.

Also if $\nabla f(\mathbf{x}^*)\mathbf{d} = 0$, then $d_2 = 0$, and for all such $\mathbf{d}$

$$\mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} = 2d_1^2 \geq 0.$$

Hence the second order necessary conditions for $\mathbf{x}^*$ to be locally minimum is also satisfied.

One can easily check that f has a global minimum at $x_1 = \frac{1}{2}, x_2 = 0$.

Sufficient conditions for a local minima :

Theorem 4 : Let $f : \Omega \rightarrow \mathbb{R}$ be a **twice continuously differentiable** function. Let $\mathbf{x}^*$ be an interior point of Ω . If $\mathbf{x}^*$ satisfies the following conditions

$$1. \nabla f(\mathbf{x}^*) = \mathbf{0}$$

$$2. \nabla^2 f(\mathbf{x}^*) \text{ is positive definite,}$$

then $\mathbf{x}^*$ is a **local minimum point** of f .

Proof: Since $\mathbf{x}^*$ is an interior point of Ω , if $\nabla f(\mathbf{x}^*) = \mathbf{0}$, then by Taylor's second order approximation formula for f near $\mathbf{x}^*$, we get

$$f(\mathbf{x}^* + t\mathbf{d}) = f(\mathbf{x}^*) + \frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} + o(t^2), \text{ for all } \mathbf{d} \in \mathbb{R}^n \text{ and all } t > 0 \text{ sufficiently small.}$$

For t small, $|\frac{t^2}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d}| > |o(t^2)|$ (if $\mathbf{d} \neq \mathbf{0}$).

Hence $f(\mathbf{x}^* + t\mathbf{d}) > f(\mathbf{x}^*)$ for all t sufficiently small. The above proof is not enough to conclude that $\mathbf{x}^*$ is a **local minimum point** of f .

The actual proof is given below but you may skip it if you find it difficult.

Optional reading:

$$f(\mathbf{x}^* + \mathbf{d}) = f(\mathbf{x}^*) + \frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} + o(\|\mathbf{d}\|^2).$$

Since $\nabla^2 f(\mathbf{x}^*)$ is real, symmetric (By spectral theorem for real symmetric matrices), there exists a real matrix U , orthogonal (that is, $U^T U = U U^T = I$) such that

$$\nabla^2 f(\mathbf{x}^*) = U^T \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_n \end{pmatrix} U, \text{ where } \lambda_i \text{'s are the eigenvalues of } \nabla^2 f(\mathbf{x}^*).$$

$$\Rightarrow \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} = \mathbf{y}^T \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_n \end{pmatrix} \mathbf{y}, \text{ where } \mathbf{y} = U\mathbf{d}.$$

$$\Rightarrow \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} = \sum_i \lambda_i y_i^2, \text{ where } y_i \text{ s the } i \text{ th component of } \mathbf{y}. \text{ (multiply and check).}$$

$$\Rightarrow \mathbf{d}^T \nabla^2 f(\mathbf{x}^*) \mathbf{d} \geq \lambda_{\min} \sum_i y_i^2 = \lambda_{\min} \mathbf{y}^T \mathbf{y} = \lambda_{\min} \mathbf{d}^T \mathbf{d}$$

$$(\mathbf{y}^T \mathbf{y} = \mathbf{d}^T U^T U \mathbf{d} = \mathbf{d}^T \mathbf{d} = \|\mathbf{d}\|^2 \text{ since } U^T U = I).$$

$$\text{Hence } f(\mathbf{x}^* + \mathbf{d}) \geq f(\mathbf{x}^*) + \lambda_{\min} \|\mathbf{d}\|^2 + o(\|\mathbf{d}\|^2).$$

For $\|\mathbf{d}\| > 0$ sufficiently small we have

$$|\lambda_{\min} \|\mathbf{d}\|^2| > |o(\|\mathbf{d}\|^2)| \text{ hence } f(\mathbf{x}^* + \mathbf{d}) > f(\mathbf{x}^*) \text{ for all } \|\mathbf{d}\| > 0, \text{ sufficiently small.}$$

Remark: Since **maximizing** f is same as **minimizing** $-f$, all the previous theorems have corresponding analogues for a **maximization problem** with some obvious changes. For example $\leq$ conditions in the results are replaced by $\geq$ conditions and with positive semidefinite (or positive

definite) matrices in the results are appropriately replaced by negative semidefinite matrices (or negative definite matrices).

Definition 4: A real valued function f defined on a **convex set** $\Omega \subseteq \mathbb{R}^n$ is said to be a **convex function** on Ω if for all $\mathbf{x}, \mathbf{y} \in \Omega$ and all $0 \leq \alpha \leq 1$,
 $f(\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}) \leq \alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y})$.

Definition 5: An $f : \Omega \rightarrow \mathbb{R}$ is said to be a concave function if $-f$ is a convex function.

Theorem 1 : If f is a **convex function** on Ω (a convex set), then the set $S = \{\mathbf{x} \in \Omega : f(\mathbf{x}) \leq c\}$ is a convex set (for all real c).

Proof: Exercise.

Theorem 2: Let f be a **continuously differentiable** function defined on a convex set, $\Omega \subseteq \mathbb{R}^n$, then f is **convex** on Ω if and only if $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in \Omega$.

Proof: Let $f : \Omega \rightarrow \mathbb{R}$ be a convex function. Then for all $\mathbf{x}, \mathbf{y} \in \Omega$ and all $0 \leq \alpha \leq 1$,
 $f(\alpha\mathbf{y} + (1 - \alpha)\mathbf{x}) \leq \alpha f(\mathbf{y}) + (1 - \alpha)f(\mathbf{x})$.

For all $\alpha > 0$, sufficiently small,

$$\frac{f(\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\alpha} \leq f(\mathbf{y}) - f(\mathbf{x})$$

Letting $\alpha \rightarrow 0$ we get

$$\nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x}).$$

To show the converse,

let $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in \Omega$. (**)

Fix $\mathbf{x}, \mathbf{y} \in \Omega$, and let $\mathbf{z}$ be a point in between and on the straight line segment joining $\mathbf{x}$ and $\mathbf{y}$.

That is $\mathbf{z} = \alpha\mathbf{x} + (1 - \alpha)\mathbf{y}$ for some $0 \leq \alpha \leq 1$. From (**) we get

$$f(\mathbf{x}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})(\mathbf{x} - \mathbf{z}) \text{ and}$$

$$f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})(\mathbf{y} - \mathbf{z}).$$

By multiplying the first equation by α , the second by $(1 - \alpha)$ and adding the two equations we get,

$$\alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})(\alpha\mathbf{x} + (1 - \alpha)\mathbf{y} - \mathbf{z}).$$

Since $\mathbf{z} = \alpha\mathbf{x} + (1 - \alpha)\mathbf{y}$ we get

$$\alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}) \geq f(\alpha\mathbf{x} + (1 - \alpha)\mathbf{y}).$$

But the above condition is difficult to verify. The following result gives an easy way to the convexity of a function provided that the function satisfies certain conditions.

The proof of the theorem is however **optional** reading.

Theorem 3 : Let f be a **twice continuously differentiable** function on a convex set Ω (let Ω be open or such that it has atleast one interior point), then f is convex on Ω if and only if for all $\mathbf{x} \in \Omega$, $\nabla^2 f(\mathbf{x})$ is **positive semidefinite**.

Proof: Let us assume that Ω is an open set.

Only if part:

Let f be a convex function.

Since Ω is open, for any $\mathbf{d} \in \mathbb{R}^n$ for t sufficiently small $\mathbf{x}^* + t\mathbf{d} \in \Omega$. Since f is twice continuously differentiable so by Taylor's theorem,

$$f(\mathbf{x}^* + t\mathbf{d}) = f(\mathbf{x}^*) + t\nabla f(\mathbf{x}^*)\mathbf{d} + \frac{t^2}{2}\mathbf{d}^T \nabla^2 f(\mathbf{x}^*)\mathbf{d} + o(t^2), \quad t > 0 \text{ sufficiently small.}$$

Since f is continuously differentiable and convex

$$f(\mathbf{x}^* + t\mathbf{d}) \geq f(\mathbf{x}^*) + t\nabla f(\mathbf{x}^*)\mathbf{d}$$

which implies $\frac{t^2}{2}\mathbf{d}^T\nabla^2 f(\mathbf{x}^*)\mathbf{d} + o(t^2) \geq 0$, for $t > 0$ sufficiently small.

By dividing by t^2 and taking the limit as $t \rightarrow 0$, we get $\mathbf{d}^T\nabla^2 f(\mathbf{x}^*)\mathbf{d} \geq 0$.

Since $\mathbf{d}$ was arbitrary the result follows.

If part: Let f satisfy the condition that for all $\mathbf{x} \in \Omega$, $\nabla^2 f(\mathbf{x})$ is **positive semidefinite**.

Let $\mathbf{x}, \mathbf{y} \in \Omega$, then by the mean value theorem, (you may not be aware of this theorem)

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}) + \frac{1}{2}(\mathbf{y} - \mathbf{x})^T \nabla^2 f(\mathbf{x}^*)(\mathbf{y} - \mathbf{x})$$

where $\mathbf{x}^* = \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$, $0 \leq \lambda \leq 1$.

Since for all $\mathbf{x} \in \Omega$, $\nabla^2 f(\mathbf{x})$ is **positive semidefinite**, the above implies

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x}), \text{ for all } \mathbf{x}, \mathbf{y} \in \Omega,$$

which implies, f is convex.

Revisiting Example 1 : Let $f(x_1, x_2) = x_1^3 - x_1^2 x_2 + 2x_2^2$ be defined on $\Omega = \{[x_1, x_2]^T : x_1 \geq 0, x_2 \geq 0\}$.

$$\text{Since } \nabla^2 f(\mathbf{x}) = \begin{pmatrix} 6x_1 - 2x_2 & -2x_1 \\ -2x_1 & 4 \end{pmatrix}, \quad \text{at } x_1 = 1, x_2 = 3, \quad \nabla^2 f(\mathbf{x}) = \begin{pmatrix} 0 & -2 \\ -2 & 4 \end{pmatrix}$$

is clearly not positive semidefinite. Hence f is not a convex function on Ω .

Theorem 4: Let $f : \Omega \rightarrow \mathbb{R}$ be a **continuously differentiable** function. If f is **convex** on Ω , then $\mathbf{x}^*$ is a **global minimum** of f if and only if for all **feasible directions** $\mathbf{d}$ at $\mathbf{x}^*$ $\nabla f(\mathbf{x}^*)\mathbf{d} \geq 0$.

Proof: Since the **only if** part is already shown before, we have to only show the **if** part.

Let $\mathbf{x}^* \in \Omega$ satisfy $\nabla f(\mathbf{x}^*)\mathbf{d} \geq 0$ for all feasible $\mathbf{d}$ at $\mathbf{x}^*$.

Since f is convex $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in \Omega$.

(1)

Hence for all $\mathbf{y} \in \Omega$, $f(\mathbf{y}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)(\mathbf{y} - \mathbf{x}^*)$.

Since Ω is convex, $\mathbf{x} = \alpha\mathbf{y} + (1 - \alpha)\mathbf{x}^* = \mathbf{x}^* + \alpha(\mathbf{y} - \mathbf{x}^*)$ belongs to Ω , for all $0 \leq \alpha \leq 1$, hence $(\mathbf{y} - \mathbf{x}^*)$ is a feasible direction at $\mathbf{x}^*$ and $\nabla f(\mathbf{x}^*)(\mathbf{y} - \mathbf{x}^*) \geq 0$.

Hence $f(\mathbf{y}) \geq f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)(\mathbf{y} - \mathbf{x}^*) \geq f(\mathbf{x}^*)$.

Since $\mathbf{y} \in \Omega$ was arbitrary, $\mathbf{x}^*$ is a global minimum of f .

Corollary 4: Let $f : \Omega \rightarrow \mathbb{R}$ be a **continuously differentiable** function. If f is **convex** on Ω and $\mathbf{x}^*$ an **interior point** of Ω , then $\mathbf{x}^*$ is a **global minimum** for f if and only if $\nabla f(\mathbf{x}^*) = \mathbf{0}$.

Proof: Follows from the previous result.

That the above result is not necessarily true if f is not convex as you have already seen in **Example 1**.

Remark : Since **minimizing** f is same as **maximizing** $-f$, all the previous theorems for **minimizing a convex function** have corresponding analogues for **maximizing a concave function**.

Theorem 5: Let $f : \Omega \rightarrow \mathbb{R}$ be a **convex function**, then the following statements are true.

1. Let S be the collection of all $\mathbf{x}$'s where f attains its **minimum value** (that is, the set of all optimal solutions of f for a minimization problem). Then S is a **convex set**, or in other words the set $S = \{\mathbf{x} : f(\mathbf{x}) \leq f(\mathbf{y}) \text{ for all } \mathbf{y} \in \Omega\}$, is a **convex set**.
2. If $\mathbf{x}^*$ is a **local minimum** point of f then it is also a **global minimum** point of f .

Proof:

1. If f does not have a minimum then the above result is vacuously true.

If f takes a minimum value then let $a = \min_{\mathbf{x} \in \Omega} \{f(\mathbf{x})\}$.

Since f is a convex function, $S_1 = \{\mathbf{x} : f(\mathbf{x}) \leq a\}$ is convex, by Theorem 1.

Note that $S_1 = S$.

2. To show that a local minimum of f is a global minimum of f .

If not, then let $\mathbf{x}^*$ be a local minimum point of f and let there exist a $\mathbf{y} \in \Omega$ such that $f(\mathbf{y}) < f(\mathbf{x}^*)$.

Join $\mathbf{x}^*$ and $\mathbf{y}$ by a straight line.

Since Ω is a convex set, the straight line segment joining $\mathbf{x}^*$ and $\mathbf{y}$ lies entirely in Ω .

Since f is a convex function, for all $0 < \alpha \leq 1$,

$$f((1 - \alpha)\mathbf{x}^* + \alpha\mathbf{y}) \leq (1 - \alpha)f(\mathbf{x}^*) + \alpha f(\mathbf{y}) < f(\mathbf{x}^*).$$

This contradicts that $\mathbf{x}^*$ is a local minimum point of f .

Remark: A natural question would be whether the conclusions of Theorem 5 holds good when **maximizing** a **convex** function. The answer however is a **NO**.

Take $f(x) = x^2$, $-1 \leq x \leq 2$.

From previous discussions however it is clear that the above result is true if you are **maximizing** $-f$ or **maximizing** a **concave** function.

Remark: We had seen while **minimizing** or **maximizing** a **linear** function over a **polyhedral** set, the extremum was attained in at least one **extreme point**. An extreme point of a polyhedral set, (more generally convex set) is one which cannot be written as a strict convex combination of two distinct points of that set).

But in the problem of **minimizing** a **convex** function over a **polyhedral** set, the only minimum may be attained at an **interior point** of Ω .

For example take $f(x) = x^2$, $-1 \leq x \leq 1$, the only minimum of this function on the interval $[-1, 1]$ which is a convex set is attained at $x = 0$ which is an interior point of $[-1, 1]$.

However when maximizing a convex function over a closed and bounded convex set the maximum is attained at an extreme point as the following theorem will illustrate.

The proof of the following theorem is **optional** reading.

Theorem 6: Let f be a **convex** function defined on a **closed** and **bounded convex** set Ω (so it has atleast one extreme point), if f has a global maximum in Ω then there exists an **extreme point** of Ω , where f takes its **maximum** value.

Proof: First we show that a global maximum is attained at a **boundary point** of Ω .

Let $\mathbf{x}^*$ be a global maximum of f in Ω . If $\mathbf{x}^*$ is a boundary point, then done. If not let L be a line passing through $\mathbf{x}^*$ and let the boundary points of Ω which L cuts be $\mathbf{y}_1, \mathbf{y}_2$.

Then $\mathbf{x}^* = \lambda\mathbf{y}_1 + (1 - \lambda)\mathbf{y}_2$, $0 < \lambda < 1$.

Since f is convex

$$f(\mathbf{x}^*) \leq \lambda f(\mathbf{y}_1) + (1 - \lambda)f(\mathbf{y}_2),$$

which implies $f(\mathbf{x}^*) \leq \max \{f(\mathbf{y}_1), f(\mathbf{y}_2)\}$.

Since $\mathbf{x}^*$ is a global maximum of f , atleast one of the boundary points $\mathbf{y}_1, \mathbf{y}_2$ must also be a global maximum of f , so the first part of the proof is done.

If this global maximum is also an extreme point of Ω then done, so assume that it is not an extreme point of Ω .

Let us again denote this boundary global maximum point of Ω as $\mathbf{x}^*$.

Consider a hyperplane $H_{\mathbf{x}^*} = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = d\}$, $d \in \mathbb{R}$, which passes through $\mathbf{x}^*$ and puts all the points of Ω in the half space, say, $\{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \leq d\}$ and let $S = \Omega \cap H_{\mathbf{x}^*}$.

Then $\mathbf{x}^* \in S$ is a global maximum of S and S is a closed, bounded and convex set.

Also the dimension of S is atleast one less than the dimension of Ω (you have intuitive understanding of dimension).

By the previous argument there exists a boundary point of S which is the global maximum of S , call it $\mathbf{x}^{**}$. If $\mathbf{x}^{**}$ is an extreme point of S then it can be shown that $\mathbf{x}^{**}$ is also an extreme point of Ω .

To see this, let $\mathbf{x}^{**} = \lambda \mathbf{z}_1 + (1 - \lambda) \mathbf{z}_2$, where $0 < \lambda < 1$,

and $\mathbf{z}_1, \mathbf{z}_2$ are distinct points of Ω .

Then $d = \mathbf{a}^T \mathbf{x}^{**} = \lambda \mathbf{a}^T \mathbf{z}_1 + (1 - \lambda) \mathbf{a}^T \mathbf{z}_2$.

Since every point of Ω lies in the half space $\{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \leq d\}$ and $0 < \lambda < 1$,

$d = \mathbf{a}^T \mathbf{z}_1 = \mathbf{a}^T \mathbf{z}_2$, which implies that $\mathbf{z}_1, \mathbf{z}_2$ are distinct points of S ,

which contradicts that $\mathbf{x}^{**}$ is an extreme point of S .

So if $\mathbf{x}^{**}$ is an extreme point of S , then $\mathbf{x}^{**}$ is also an extreme point of Ω and we are done. If $\mathbf{x}^{**}$ is not an extreme point of S , then repeat the above process with $\mathbf{x}^*$ replaced by $\mathbf{x}^{**}$ and Ω replaced by S (dimension of S is less than that of Ω). After a finite number of steps the process will terminate with an extreme point of Ω . In the worst case it may terminate with S being a single point.